Exact PTO Equivalence Decimal Addition

Metric-Extracted Controller

`analyses/metric_extraction/DFST/add10/warmup_50000_steps_4000000/`

1 Statement and Scope

Controllers. The exported table `minimized_transition_table.pdf` defines controller C, with initial state 0 and halt state 51. The function `add_grid(10)` defines expert $\mathcal{E}$, with initial state 0 and halt state 18. The adjacent expert export matches this function entry for entry. All tables retain the exported state numbers.

Input Domain. Let $D = \{0, 1, \ldots, 9\}$. Row 0 initially contains the contiguous word $a + b$, starting at column 0, with $a \in D^m$, $b \in D^n$, and $m, n \geq 1$. All remaining cells contain blanks. The head starts at $(0, 0)$. Write $A = \mathrm{val}_{10}(a)$ and $B = \mathrm{val}_{10}(b)$. The domain permits arbitrary operand lengths, unequal lengths, and leading zeros.

Execution Convention. The grid uses $\mathbb{Z}^2$, with coordinates increasing rightward and downward. The symbol λ denotes a blank. A transition (q', w, M) writes w, performs motion M, and enters state q'. Motions L, R, D, U move one cell. Motion S writes and stops without moving.

Policy-Trajectory Observations. For input $a + b$, define the policy-trajectory observation (PTO) by

$$\mathrm{PTO}(a + b) = \big((r_t, w_t, M_t)\big)_{t=0}^{T-1},$$

where r_t, w_t, and M_t denote the read symbol, written symbol, and motion at controller step t. A finite execution contributes the final S action exactly once, without halt-loop padding.

Theorem 1 (Exact PTO Equivalence)**.** *For every input in the stated domain,* $\mathrm{PTO}_C(a + b) = \mathrm{PTO}_{\mathcal{E}}(a + b)$. *Both executions terminate on the same step. The complete grids and head positions agree at every step, including the final configuration.*

The theorem concerns autonomous executions of the exported discrete controller on the stated input domain.

2 Expert Behavior

2.1 Accumulator and Marker Invariant

Write $B = \sum_{j=0}^{n-1} b_j 10^j$, with b_0 the rightmost digit. The expert scans to the separator at column m, steps left and down, writes a working + below column $m-1$, and enters state 16 above the marker.

Lemma 1 (Round Invariant)**.** *After processing k right-operand digits, $0 \le k \le n$, the expert enters state* 16 *at $(p_k, 0)$, where $p_k = m - 1 - k$. At this point:*

- *Row* 1 *contains exactly one nonblank symbol: a + at column p_k.*
- *Row* 0 *contains a contiguous accumulator on $[\ell_k, m-1]$, with value $C_k = A + \sum_{j<k} b_j 10^j$ and left edge $\ell_k \le p_k + 1$. Column $m-1$ carries the units digit.*
- *The separator remains at column m. The unprocessed prefix of b occupies columns $m+1, \ldots, m+n-k$. All remaining cells contain blanks.*

Proof. Initialization gives $k = 0$, $\ell_0 = 0$, $C_0 = A$, and $p_0 = m - 1$. State 16 preserves the digit at p_k, or writes 0 on a blank. A blank at p_k forces $\ell_k = p_k + 1$, so zero extension preserves contiguity.

For $k < n$, the right scan reaches the blank after the unprocessed prefix, steps left to digit b_k at $r_k = m + n - k$, erases b_k, and descends. The expert transports b_k left along row 1 to the marker. The distance satisfies

$$r_k - p_k = n + 1 \ge 2. \tag{1}$$

Consequently, the first two transport steps read blanks. At the marker, the expert ascends and adds b_k to the accumulator digit e at column p_k. If $e + b_k < 10$, the expert writes $e + b_k$ and descends. Otherwise, the expert writes $e + b_k - 10$, moves left, and propagates one carry. Each digit 9 changes to 0; the first smaller digit increases by one, or a blank changes to 1. The update adds exactly $b_k 10^k$ and preserves contiguity.

The expert returns along row 1, erases the marker, steps left, writes the next marker at $p_{k+1} = p_k - 1$, and ascends into state 16. The earlier zero extension gives $\ell_{k+1} \le p_k = p_{k+1} + 1$. The accumulator now has value C_{k+1}, establishing the next round. □

Lemma 2 (Termination and Final Configuration)**.** *The expert terminates with only the padded base-*10 *representation of $A + B$ on row* 0*, ending at column $m - 1$. The final head stops at $(m - n - 1, 0)$.*

Writing $\ell_{10}(s)$ for the decimal digit count, with $\ell_{10}(0) = 1$, the output width equals $L = \max\{m, n+1, \ell_{10}(A+B)\}$.

Proof. Each round consumes one of the n right-operand digits. Every scan crosses a finite interval, and every carry reaches a digit below 9 or a blank. After round n, state 16 ensures a digit at p_n. The next right scan returns to the separator. State 3 erases the separator and descends. State 17 walks left, erases the final marker, ascends, preserves the accumulator digit, and stops. The invariant gives $C_n = A + B$, and cleanup removes all work symbols.

The original m columns persist, and marker extension reaches p_n, ensuring at least $n+1$ digits. Every additional carry extension contributes a digit required by the current numerical value. Since partial sums never decrease, the final width equals L. □

3 Observation Protocol

The observation protocol refines expert states by geometric context. Each phase records the permitted read symbols and successor phases. Section 4 gives the complete protocol together with the controller-state relation; the shared tables avoid repeating the same phase definitions.

Lemma 3 (Protocol Coverage)**.** *Every expert execution on a valid decimal addition grid follows the* 59 *phases in Section 4, starting at phase* 0 *and ending at phase* 58*.*

Proof. Initialization satisfies the first phases because both operands contain a digit. Lemma 1 gives the contiguous accumulator, the ensured digit, the single working marker and the remaining right-operand prefix in every round. The explanations in Sections 4.1 and 4.2 establish all first-cell, boundary, transport and return readings. The carry rule processes exactly the digits permitted in phase 50. Induction over the transitions of each round preserves these conditions. Lemma 2 establishes finite cleanup and the final phase. □

4 Simulation Certificate

4.1 Scanning, Carrying, and Cleanup

A proof phase records geometric context. Phase i pairs expert state g_i with controller state $q \in Q_i$, with identical grids and heads. An entry $x : (j, w, M)$ means read x, write w, move M, and advance to phase j. The index j denotes a phase. Each Q_i lists the related controller states.

Here $e \in D$ and $h \in \{0, \ldots, 8\}$. A digit denotes the corresponding symbol when read or written. Phases 10–49 comprise ten digit-transport families in the next certificate subsection, with $F_d = 10 + 4d$.

i	Phase	g_i	Q_i	Read : (next phase, write, move)
0	First left-operand digit	0	$\{0\}$	$e : (1, e, \mathsf{R})$
1	Find the input separator	0	$\{0\}$	$e : (1, e, \mathsf{R})$ $+ : (2, +, \mathsf{L})$
2	Descend below units digit	1	$\{1\}$	$e : (3, e, \mathsf{D})$
3	Place the working marker	1	$\{2\}$	$\lambda : (4, +, \mathsf{U})$
4	Ensure a digit above marker	16	$\{3\}$	$e : (6, e, \mathsf{R})$ $\lambda : (5, 0, \mathsf{R})$
5	Adjacent existing digit	2	$\{4\}$	$e : (6, e, \mathsf{R})$
6	Scan accumulator to separator	2	$\{4\}$	$e : (6, e, \mathsf{R})$ $+ : (7, +, \mathsf{R})$
7	Inspect remaining right operand	2	$\{5\}$	$e : (8, e, \mathsf{R})$ $\lambda : (54, \lambda, \mathsf{L})$
8	Scan remaining right operand	2	$\{5\}$	$e : (8, e, \mathsf{R})$ $\lambda : (9, \lambda, \mathsf{L})$
9	Read rightmost factor digit	3	$\{6\}$	$d : (F_d, \lambda, \mathsf{D})$ $(d \in D)$
50	Propagate decimal carry	14	$\{11\}$	$9 : (50, 0, \mathsf{L})$ $h : (51, h + 1, \mathsf{D})$ $\lambda : (51, 1, \mathsf{D})$
51	First return cell after carry	15	$\{15\}$	$\lambda : (52, \lambda, \mathsf{R})$
52	Return to working marker	15	$\{15\}$	$\lambda : (52, \lambda, \mathsf{R})$ $+ : (3, \lambda, \mathsf{L})$
53	Erase and shift marker	15	$\{11, 15\}$	$+ : (3, \lambda, \mathsf{L})$
54	Erase input separator	3	$\{6\}$	$+ : (55, \lambda, \mathsf{D})$
55	Begin cleanup below separator	17	$\{24\}$	$\lambda : (56, \lambda, \mathsf{L})$
56	Erase final working marker	17	$\{25\}$	$\lambda : (56, \lambda, \mathsf{L})$ $+ : (57, \lambda, \mathsf{U})$
57	Preserve final digit and stop	17	$\{26\}$	$e : (58, e, \mathsf{S})$
58	Halted	18	$\{51\}$	–

Why These Read Symbols Cover Every Valid Run. Both operands contain at least one digit. Initialization therefore reads a digit before the separator, descends below a digit and places the marker on a blank. Lemma 1 places an existing digit immediately after each zero extension. The scan crosses the contiguous accumulator to the separator. Immediately to the separator's right, a blank means the next left step reads that separator; after any digit, the next left step instead reads a digit.

A carry moves strictly left of the marker, so the first return cell on row 1 contains a blank. A noncarry addition descends directly onto the marker. Erasing and shifting the marker reaches a blank. Cleanup starts below the separator on a blank, walks to the marker, then ascends to a digit. These facts establish the table's restrictions.

4.2 Digit-Transport Phases

For each $d \in D$, four phases carry the selected digit to the accumulator:

$$F_d = 10 + 4d, \qquad H_d = F_d + 1, \qquad W_d = F_d + 2, \qquad A_d = F_d + 3.$$

All four phases use expert state $4 + d$, paired with singleton controller sets containing f_d, h_d, w_d, a_d, respectively. The protocol follows:

Phase	Meaning	Read : (next phase, write, move)
F_d	First transport blank	$\lambda : (H_d, \lambda, \mathsf{L})$
H_d	Second transport blank	$\lambda : (W_d, \lambda, \mathsf{L})$
W_d	Walk left to the marker	$\lambda : (W_d, \lambda, \mathsf{L}) \quad + : (A_d, +, \mathsf{U})$
A_d	Add at the marked column	$e : (53, d + e, \mathsf{D})$ if $d + e < 10$
		$e : (50, d + e - 10, \mathsf{L})$ if $d + e \geq 10$

The Actual Controller-State Numbers. Each row retains the state numbers from the exported controller table. This table expands the entire 40-phase family.

d	Expert $4 + d$	f_d	h_d	w_d	a_d
0	4	27	28	29	30
1	5	12	13	14	11
2	6	39	40	41	42
3	7	31	32	33	34
4	8	43	44	45	46
5	9	47	48	49	50
6	10	7	8	9	10
7	11	20	21	22	23
8	12	16	17	18	19
9	13	35	36	37	38

Transport Coverage. Equation (1) gives $r_k - p_k = n + 1 \geq 2$. The first two transport cells therefore contain blanks. After two left steps, the head reaches either the marker or a blank to the marker's right. The walk reaches the marker and ascends to a digit ensured at the start of the round. The two branches implement decimal addition with at most one carry:

$$d + e = s + 10c, \qquad s \in D, \quad c \in \{0, 1\}.$$

This accounts for every pair (d, e), including 0, 9 and all carry cases.

Multiple Roles of Controller State 11. State 11 performs addition of 1 when reading a digit, propagates carries when entered to the left of an overflowing column, and erases the marker when reading +. With $d = 1$ and no carry, the controller enters state 15; every other noncarry branch enters state 11. Both belong to $Q_{53} = \{11, 15\}$ and have the same required action on +. Every overflow enters state $11 \in Q_{50}$. The phase relation distinguishes these geometric contexts and therefore does not require a one-to-one correspondence between controller states.

Local Transition Identity. For every phase i, controller state $q \in Q_i$ and listed branch $x : (j, w, M)$, direct comparison with the exported tables gives

$$\delta_{\mathcal{E}}(g_i, x) = (g_j, w, M), \qquad \delta_C(q, x) = (q', w, M), \quad q' \in Q_j. \tag{2}$$

Every required entry exists. A transition reaches a halt state exactly when $j = 58$, with $M = S$. The certificate contains 60 phase–state pairs, including the terminal pair, and 266 local checks of (2). The checks cover all 223 defined nonterminal controller entries. The remaining 12 entries contain halt loops, excluded from PTOs.

5 Equivalence Proof

Proof of Theorem 1. Initially both configurations have the same grid and head, with control states 0 related by phase 0. Suppose the relation holds before a controller step. Lemma 3 supplies an allowed read symbol x. Equal grids and heads give the same read symbol to both controllers. Equation (2) gives equal writes and motions and preserves the relation between successor states. The next grids and heads therefore agree.

Induction establishes equality of every finite execution prefix. The terminal condition permits a halt exactly on an edge to the terminal phase. Lemma 2 establishes termination. Both controllers therefore halt together with equal complete PTOs and final configurations. □

Protocol coverage and the local transition identity exclude every undefined entry on the stated input domain.

Input	Final digit string	Occupied columns	Final head
$0 + 0$	00	$[-1, 0]$	$(-1, 0)$
$9 + 9$	18	$[-1, 0]$	$(-1, 0)$
$999 + 1$	1000	$[-1, 2]$	$(1, 0)$
$0 + 999$	0999	$[-3, 0]$	$(-3, 0)$
$001 + 09$	010	$[0, 2]$	$(0, 0)$

6 Verification and Artifacts

The universal claim follows from the geometric invariant, protocol coverage and induction. The following checks bind the finite certificate to the exact adjacent files and independently exercise the controller.

Finite Certificate. Run

```
python3 verify_equivalence.py --quick
```

from this proof directory. This uses only the Python standard library and checks:

- SHA-256 fingerprints of the controller exports, the expert-source snapshots and the finite protocol;
- exact agreement between the minimized CSV and JSON controllers;
- exact agreement between the exported expert and `add_grid(10)`;
- all 266 local obligations for the 60 related phase–state pairs, including start, halt and successor closure;
- coverage of all 223 defined nonterminal controller entries.

The file `equivalence_verification.json` records the phase relation and every checked transition.

Independent Complete Executions. Omitting `--quick` also runs two separate grid memories and compares every read symbol, write, move and head position. The checker compares the complete final grids against the exact padded decimal sum and final-head formula. The validation set contains all 12,100 pairs of operand strings of lengths 1–2, plus 44 directed cases and 256 fixed-seed random cases. Directed cases include long all-9 carry chains, zeros, unequal lengths, leading zeros and operands of length 1024.

All 3,577,109 compared PTO steps agree. The execution checks exercise every defined nonterminal controller entry. The finite tests supplement the length-independent proof.

Printed-Table Verification. Printed-table verification compares all 624 cells of the two-page PDF against the CSV: 235 defined entries and 389 undefined cells. The optional command

```
python3 verify_equivalence.py --quick --check-pdf
```

repeats this printed-table check with `pypdf` installed.

Artifact Identity. The title page gives the repository-relative case directory. This proof directory contains the controller and expert exports, reference code, protocol, checker, and saved verification report. The verification script stores or reads SHA-256 checksums for these artifacts. A checksum identifies the exact bytes of a file; the checker compares the stored and computed values to detect changes. These checks bind the certificate to the supplied exports. The invariants, protocol coverage, and local transition identity establish equivalence.